



QUIZ MASTER

EFFICIENT QUIZ PLATFORM

Smart Learning, Smarter Testing

A PROJECT REPORT FOR

QUIZ MASTER APPLICATION

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I'm currently pursuing the IITM BS program in Data Science and Programming. I'm super passionate about tech and love building things—whether it's web apps, data-driven projects, or just experimenting with new ideas. I'm always up for a challenge and enjoy learning something new along the way. Excited about connecting with like-minded people and exploring the endless possibilities in tech.x

Description:

This project is built as part of the Modern Application Development - I course from the IITM BS Degree program.

It's a quiz-based learning platform designed for both students and administrators.

- Admins have full control over the content—they can create, edit, and manage subjects, chapters, questions, and quizzes to keep the platform updated and engaging.
- Users (students or learners) can take quizzes, track their past performances, and explore subject details, making learning more interactive and structured.

The platform is built with scalability in mind, meaning it can grow with new features like analytics, adaptive learning, and interactive tools in the future. Whether you're an admin curating quizzes or a learner testing your knowledge, this app makes the process smooth and efficient!

Technologies used:

Below are some of the important backend modules and libraries that I've used in this project.

- **Flask** – The core framework, managing web requests and application logic.
- **Jinja Templating** – Dynamically generates HTML pages, integrating seamlessly with Flask.
- **Flask_SQLAlchemy** – An ORM that simplifies database interactions using Python objects.
- **Requests** – Simplifies API interactions for data exchange.
- **Tools & IDE:** Visual Studio Code, SQLite Editor

Frontend Technologies

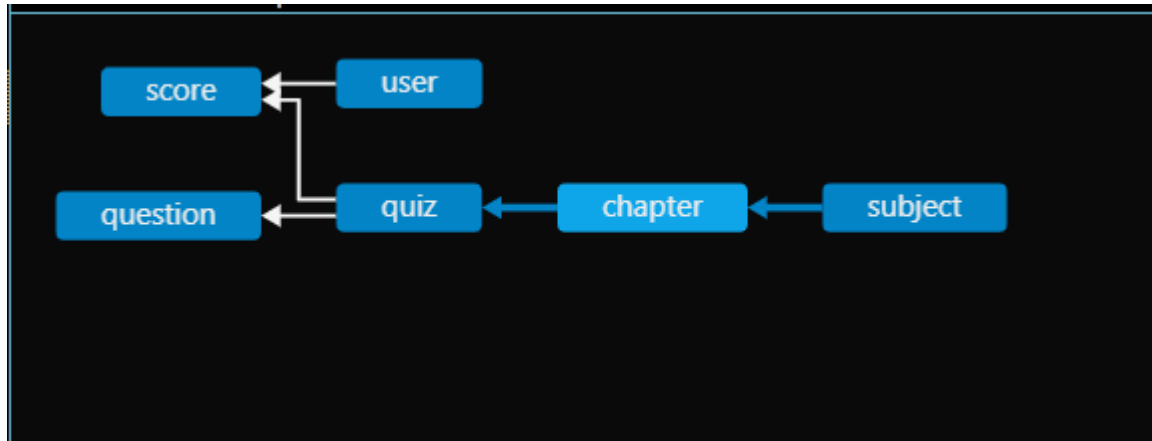
The frontend is designed with a clean and responsive interface, using:

- **HTML, CSS, JavaScript** – Core web technologies for structuring and styling the user interface.
- **Bootstrap (Latest Version)** – A modern, flexible framework with pre-styled components and a responsive grid system.
- **Chart.js** – A JavaScript library used for interactive and customizable data visualizations

In addition to these, several other modules and libraries contribute to the functionality and security of this web application.

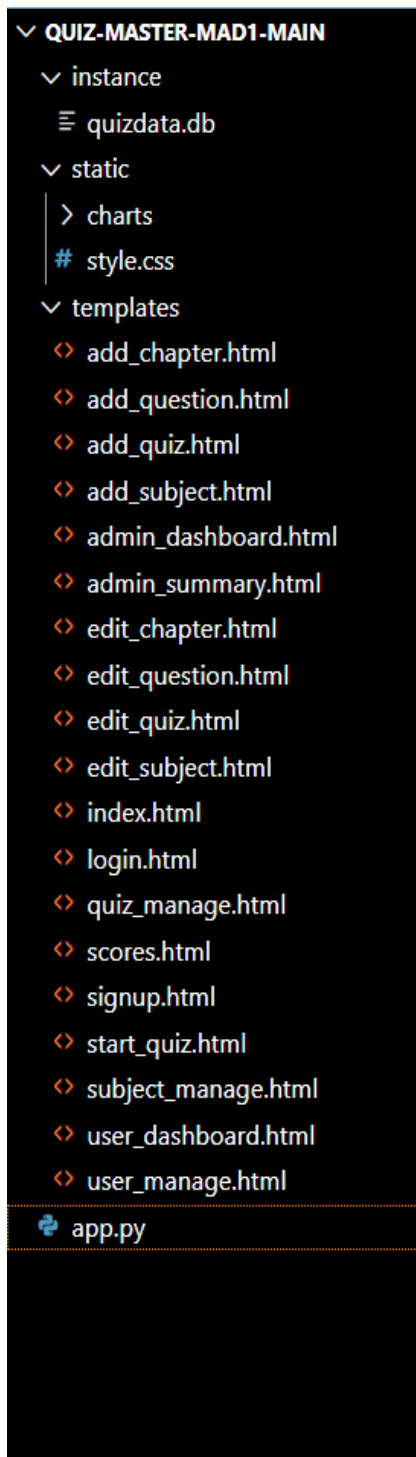
Database Schema Design:

This application uses a relational database with 6 tables, ensuring data integrity, role-based access, and efficient record tracking using foreign keys and timestamps. The schema is normalized, eliminating multi-valued and transitive dependencies.



- User: Stores user information (username, fullname, DOB, qualification, password).
- Subject: Stores subjects with descriptions.
- Chapter: Stores different chapters under subjects
- Question: Stores questions, options, and correct answers.
- Quiz: Stores quiz details such as date and time.
- Scores: Stores user quiz scores and performance data.

Architecture and Features:



- ➔ static This folder is used to store static files such as CSS, JavaScript, and images
- ➔ style.css This is a CSS file that likely contains the styling for the application's front-end.
- ➔ Templates The **templates/** folder contains multiple HTML files, indicating the use of Flask or Django's templating system.
- ➔ app.py Handles routing, request handling, and connects to the database.

Our multi-user quiz app offers a seamless experience for both admins and users. Admins can manage subjects, quizzes, questions, and student details, with graphs and transcripts for performance tracking. Users can explore subjects, attempt live quizzes with timers, review past performances, and send queries via the contact page. They can also edit profiles and upload profile photos for a personalized experience. With search functionality, secure logout, and smart learning features, the platform ensures an efficient and engaging quiz experience.

Video:

https://drive.google.com/file/d/1Qsr4hS3DgZYzPBpXZCgV_nSjQGikFNlz/view?usp=drive_link